

Project Progress Report: Updated Content

Paper Summary

Introduction

Skin lesion classification from dermoscopic images can help organize visual evidence for clinical review, but similar appearance across diagnoses and severe class imbalance make the task difficult. This project uses the HAM10000 seven-class dataset and investigates an EfficientNet-based image classifier enhanced with CBAM attention and Grad-CAM explanation. The implementation and several validation experiments are now complete. The original lesion-disjoint test partition remains reserved for final assessment.

The present report updates the earlier research and planning stage. It records the fixed data split, trained baseline and alternatives, feature and inference studies, the later protocol exploration, and current best validation results. Exploratory image-level and lesion-disjoint scores are presented separately because they have different evaluation properties. A final test-set result is not yet available.

The project is image-only by design. Age, sex, and lesion location were examined only in an auxiliary feasibility probe and were not adopted as final inputs. This keeps the principal research question focused on visual representation, attention, explanation, and a transparent evaluation protocol. The local implementation retains configurations and checkpoints so that each reported validation figure can be traced to saved artifacts.

Literature Review

The HAM10000 dataset paper established a large public seven-category dermoscopy collection with 10,015 images and repeated views of some lesions. The selected base paper by Pardede et al. compared EfficientNet-B0 and MobileNetV3-Large with class balancing and Grad-CAM, reporting 98.86% and 98.88% accuracy respectively. It motivated an efficient pretrained backbone, explicit class-imbalance treatment, and interpretable visualization. Its published test support totals 7,041 balanced cases, unlike an untouched 15% of original HAM10000, which would be about 1,502 images. Its exact oversampling and split sequence cannot be proven without code and image IDs, so that result is not directly comparable with this project's untouched original-image validation.

Further work influenced focused checks. Haque et al. reported 91.15% accuracy and 0.8545 macro-F1 with a much larger EfficientNetV2-L attention model at 384 pixels, motivating a low-cost resolution probe. An EfficientNet-B2 public implementation used image-level splitting and stronger augmentation; its source code informed a controlled augmentation experiment. The PanDerm dermatology-pretrained model motivated a separate feature probe. A three-category ensemble paper reported 96% on a balanced HAM10000 subset, illustrating why class count and evaluation cohort must accompany any headline percentage. These studies are research comparisons, not direct replicas of the project's strict seven-class protocol.

The literature also encouraged attention to explanations rather than accuracy alone. Grad-CAM can make spatial evidence visible, but it requires careful interpretation: a heatmap may highlight background, imaging artifacts, or non-lesion structures, and visual plausibility is not equivalent to clinical validity. The earlier report listed several explanation techniques; the implemented project selected Grad-CAM as its primary technique to keep the method coherent and computationally manageable.

Comparative Analysis of Existing Methods

Pardede et al. [2]

Dataset: HAM10000 | Method: EfficientNet-B0, oversampling, Grad-CAM

Advantage: Relevant base architecture and imbalance study

Limitation: Table IV has 7,041 balanced test cases; not an untouched original-image 15% cohort

Reported performance: 98.86% accuracy reported

Haque et al. [3]

Dataset: HAM10000 | Method: EfficientNetV2-L, attention, 384-pixel input

Advantage: Motivated resolution and attention research

Limitation: Much larger model; lesion-level separation not established

Reported performance: 91.15% accuracy and 0.8545 macro-F1 reported

PanDerm [4]

Dataset: Dermatology images / HAM10000 probe | Method: Dermatology-pretrained transformer features

Advantage: Complementary predictions in a project probe

Limitation: Outside the approved final EfficientNet architecture

Reported performance: Project strict PanDerm/B0 ensemble: 87.69% validation accuracy

Our strict weighted B0 + CBAM

Dataset: HAM10000, seven classes | Method: Lesion-disjoint split, weighted cross entropy

Advantage: Reproducible approved-family trained baseline

Limitation: Validation result; no final test evaluation

Reported performance: 84.56% accuracy, 0.7569 macro-F1

Our exploratory four B0 streams

Dataset: HAM10000, seven classes | Method: Two B0 + CBAM checkpoints at 224 and 384 pixels

Advantage: Improved exploratory inference without new training

Limitation: Shared-lesion image-level validation; not a strict result

Reported performance: 88.16% accuracy, 0.8280 macro-F1

Our exploratory B0 + PanDerm ensemble

Dataset: HAM10000, seven classes | Method: Four B0 streams plus four PanDerm orientation views

Advantage: Highest numeric validation accuracy

Limitation: Shared-lesion validation and outside final architecture

Reported performance: 90.75% accuracy, 0.8573 macro-F1

1. Skin Lesion Classification Using Deep Learning

Convolutional neural networks learn increasingly complex visual features from pixels. EfficientNet combines depth, width, and image-resolution scaling to offer a useful accuracy and compute trade-off. This project uses ImageNet-pretrained EfficientNet-B0 as the completed core backbone. Larger B variants are research possibilities, not trained results in the current record.

2. Attention Mechanisms in Skin Lesion Classification

CBAM uses learned channel and spatial attention on the final EfficientNet feature map. Channel attention reweights feature types; spatial attention reweights map locations. This addition is implemented in the core model. Attention can guide feature use but should not be described as guaranteed lesion localization or a proven reason for improvement without a matching non-CBAM ablation.

3. EfficientNet as the Proposed Backbone

EfficientNet-B0 was chosen because pretrained weights are available and its memory demand fits the available local GPU. The completed architecture consists of the B0 feature extractor, one CBAM block after its final spatial map, global average pooling, and a seven-class head. The classifier dimensions and dropout schedule are this project's implementation choices rather than claims copied from the base paper.

4. Explainable Artificial Intelligence

Grad-CAM code is implemented to generate class-specific heatmaps from a trained checkpoint. CBAM influences feature processing during inference; Grad-CAM analyzes a prediction afterward. Systematic qualitative review of correct and incorrect cases, especially minority-class errors, remains to be completed before a final interpretability claim is made.

5. HAM10000 Dataset

The original dataset has 10,015 images and 7,470 lesion IDs. Class codes are nv, mel, bkl, bcc, akiec, vasc, and df. The repository validates image paths and labels and retains a fixed stratified group split: 7,009 train, 1,503 validation, and 1,503 locked test images. No lesion ID overlaps among these original partitions. The project uses images as model inputs and lesion IDs only for safe partitioning.

The partition was produced with a fixed random seed and stratified group folds. This is stricter than simply dividing image rows at random because HAM10000 can contain more than one image of a lesion. Grouping reduces the specific risk of a model recognizing a lesion already encountered during training. It does not by itself eliminate all sources of bias,

such as acquisition differences or repeated validation selection. Those limitations are reported alongside the numbers.

Class Code	Lesion Category
nv	Melanocytic nevus
mel	Melanoma
bkl	Benign keratosis-like lesion
bcc	Basal cell carcinoma
akiec	Actinic keratosis / intraepithelial carcinoma
vasc	Vascular lesion
df	Dermatofibroma

6. Class Imbalance

The nevus class has 6,705 original images while dermatofibroma has 115. The earlier report proposed balancing before data splitting. The implemented strict protocol corrects that plan: splitting occurs first by lesion ID, and weights or repeated references are derived from training data only. Validation and test retain their original distributions. This prevents balancing from introducing duplicates across evaluation partitions.

Training-frequency-weighted cross entropy yielded 84.56% strict validation accuracy and 0.7569 macro-F1. Full random oversampling raised accuracy to 85.30% but reduced macro-F1 to 0.7321 and showed repeated-reference overfitting. Focal loss achieved 83.77% and 0.7238. The results show that imbalance handling must be judged with class-balanced metrics, not accuracy alone.

The oversampling experiment raised every training category to the original majority count through repeated references with newly sampled training transforms. It therefore exposed the model to rare examples more often without introducing duplicates into validation. The final oversampled training accuracy was much higher than its validation accuracy, a sign that repeating small minority collections did not supply independent visual diversity. Weighted loss remains an important baseline because its macro-F1 exceeded that of the oversampled and focal trained checkpoints.

7. Base Paper

Pardede et al. [2] remains the base reference for EfficientNet-B0, class balancing, conservative augmentation, and Grad-CAM in a seven-class HAM10000 task. The project extends its broad direction with CBAM and a fixed lesion-disjoint development protocol. A repository audit found that the paper's reported balanced test support cannot represent an untouched 15% sample of the original HAM10000 images. The exact processing sequence is uncertain, so the reported 98.86% is cited as the authors' number under a different or incompletely specified evaluation protocol, not as a directly comparable benchmark.

8. Proposed Methodology

The implemented core flow is HAM10000 images and metadata → fixed lesion-ID split → training-only class weights or oversampling → image resizing and normalization → ImageNet-pretrained EfficientNet-B0 → CBAM → seven-class predictions → validation metrics and Grad-CAM inspection. Training uses 224 × 224 inputs, AdamW, mixed precision, batch size 64, and macro-F1-based checkpoint selection. Each run saves its best and latest checkpoints and validation history.

The primary evaluation method is lesion-disjoint validation. A separate image-level development split was created for protocol exploration using only the original train-plus-validation pool; original test images remain excluded. Its 563 shared lesion IDs affect 596 validation images. Results on this split are labeled exploratory and are not substituted for strict or final test performance. PanDerm feature work is also labeled exploratory because it differs from the finalized EfficientNet, CBAM, Grad-CAM architecture.

On the exploratory split, the weighted B0 plus CBAM checkpoint achieved 86.23% accuracy and 0.7808 macro-F1. A stronger augmentation recipe using vertical and horizontal flips, rotation, small affine changes, and mild color jitter was tested under the same split and model. It reached 85.30% accuracy and 0.7657 macro-F1. The experiment reduced apparent memorization in training but did not improve the validation measures; that recipe was therefore not selected. These two checkpoints were preserved and later reused for an inference-only multi-resolution average.

Stage 1 — Dataset Preparation

Completed: verified original images and metadata, label mapping, lesion IDs, and fixed split assignment.

Stage 2 — Class Balancing

Completed controlled training-only weighted-loss, oversampling, and focal-loss comparisons. Weighted loss remained the better original checkpoint for balanced validation performance.

Stage 3 — Dataset Split

Completed original 7,009/1,503/1,503 lesion-disjoint train/validation/test split. The test partition remains locked. The separate image-level development split is for explicitly labeled exploratory work only.

Stage 4 — EfficientNet

Completed ImageNet-pretrained EfficientNet-B0 fine tuning and saved checkpoints. Other EfficientNet variants have not yet been established as completed trained models.

Stage 5 — Attention

Completed one post-backbone CBAM module with channel and spatial attention in the core classifier.

Stage 6 — Classification

Completed seven-class classification, saved probabilities, class metrics, confusion analyses, checkpoint selection, and inference scripts.

Stage 7 — Explainability

Grad-CAM generation code is available. A systematic explanation review and final figure selection remain pending.

9. Performance Evaluation

The project reports accuracy, per-class precision and recall, macro-F1, and confusion matrices. Macro-F1 is especially important because the seven classes vary greatly in size. Each score is tied to its split and checkpoint or ensemble. AUC-ROC was listed in the original plan but is not reported here as a completed measurement.

Protocol / method	Validation accuracy	Macro-F1	Status
Strict weighted B0 + CBAM	84.56%	0.7569	Original trained core baseline
Strict oversampled B0 + CBAM	85.30%	0.7321	Accuracy gain, class-balance loss
Strict focal B0 + CBAM	83.77%	0.7238	Did not improve leader
Strict weighted + focal ensemble	84.36%	0.7718	Best pre-exploration strict macro-F1
Strict prior-corrected oversampled B0	85.89%	0.7227	Best pre-exploration strict accuracy
Strict PanDerm + B0 ensemble, accuracy leader	87.69%	0.7771	Exploratory architecture probe
Strict PanDerm + B0, macro-F1 leader	87.03%	0.7837	Different configuration
Image-level four B0 + CBAM streams	88.16%	0.8280	Shared-lesion exploratory validation
Image-level eight B0 + PanDerm streams	90.75%	0.8573	Highest numeric validation accuracy; shared lesions

The 90.75% result is the project's highest full-validation numeric accuracy. It is not a lesion-disjoint result or a final test score. The highest strict accuracy is 87.69%; the highest strict macro-F1 is 0.7837 from another configuration. The highest EfficientNet plus CBAM exploratory result is 88.16%. No held-out test metric has been produced.

Additional no-training comparisons help explain these outcomes. On strict validation, horizontal-flip test-time averaging improved the weighted checkpoint's macro-F1 to 0.7704 while slightly reducing accuracy. The weighted-plus-focal probability average then reached 0.7718 macro-F1. A frozen classifier on post-CBAM features, metadata fusion, MC-dropout averaging, and an aspect-preserving center-crop diagnostic all fell short of the existing leaders. The metadata study did not pass its preset improvement threshold, so a costly multimodal retraining study was not initiated. Prior correction moved some predictions toward the majority class and raised strict accuracy to 85.89%, but macro-F1 fell to 0.7227; this was not selected as a balanced solution.

Multi-resolution averaging of two saved B0 plus CBAM checkpoints at both 224 and 384 pixels reached 88.16% accuracy and 0.8280 macro-F1 on exploratory image-level validation without new CNN training. A comparable strict

validation check did not surpass the previous strict leaders, limiting this finding to the exploratory cohort. PanDerm frozen features and shallow heads were then evaluated separately. A short two-epoch PanDerm fine-tuning pilot was stopped after missing its preset threshold. Although the pilot alone did not improve, PanDerm predictions complemented B0 in an ensemble. Four orientation views of the PanDerm SVM combined with B0 streams produced the 90.75% exploratory result, whereas the corresponding strict orientation ensemble reached 87.69%. This difference reinforces the need to label protocols and architectures precisely.

10. Research Gap

The project addresses four practical gaps that emerged during research. First, rare classes can be obscured by overall accuracy. Second, repeat-lesion images and balancing order can undermine interpretation of a benchmark. Third, attention and explanation are often discussed together even though they play different roles; CBAM modifies features, while Grad-CAM visualizes selected class evidence. Fourth, larger ensembles and dermatology-pretrained backbones may improve validation scores but complicate a lightweight final EfficientNet method. These issues are treated as empirical and reporting questions, not solved by one headline score.

11. Proposed Contribution

The project contributes a reproducible image-only EfficientNet-B0 plus CBAM implementation, a lesion-disjoint HAM10000 development protocol, controlled imbalance comparisons, saved checkpoint and experiment records, and a Grad-CAM path for explanation. Its separate exploratory investigations show how protocol choice and complementary pretrained features affect measured validation accuracy. The final paper will distinguish those insights from the approved EfficientNet, CBAM, and Grad-CAM method and from any later held-out test result.

12. Current Progress

- Completed dataset acquisition, metadata validation, seven-class mapping, image-path audit, and fixed lesion-disjoint split.
- Implemented and trained weighted, oversampled, and focal EfficientNet-B0 plus CBAM runs; retained histories, best/latest checkpoints, and validation metrics.
- Completed test-time flip averaging, probability ensemble, frozen-feature classifier, metadata-feasibility, MC-dropout, crop, and prior-correction analyses.
- Completed controlled exploratory image-level training and stronger-augmentation comparison; preserved both trained checkpoints.
- Achieved 88.16% exploratory validation accuracy with a four-stream saved-checkpoint EfficientNet plus CBAM ensemble.
- Completed a PanDerm research probe and an eight-stream exploratory validation ensemble that reached 90.75%; kept this outside the finalized architecture.
- Audited selected high published accuracy claims against class counts, cohorts, and split methods; documented the base-paper support inconsistency.
- Kept the original 1,503-image test partition locked; final model selection, explanation review, and final evaluation remain.

The core software and validation work are substantially complete. Current work is focused on interpreting results, selecting a method consistent with the guide-approved architecture, reviewing explanations, and writing the report and paper. No further training is being conducted for this reporting period.

Saved local checkpoints include the original weighted, oversampled, and focal B0 runs; both exploratory B0 training runs; and the PanDerm pilot. Validation metric JSON files, experiment summaries, and inference scripts support the reported comparisons. Third-party PanDerm weights and raw dataset images are not redistributed in the Git repository. This separation preserves the reproducible project code and numerical records while respecting the size and source of external artifacts.

Conclusion

Since the previous progress report, the project has moved from dataset and literature preparation to a functioning, checkpointed classifier and substantial experimental analysis. The findings show that weighted training provides a solid EfficientNet-B0 plus CBAM baseline, while alternative balancing methods and inference corrections involve accuracy-versus-macro-F1 trade-offs. PanDerm-assisted ensembles produce the highest numeric validation values but are explicitly exploratory and outside the finalized architecture. The final claim will depend on a locked configuration, Grad-CAM review, and one untouched test evaluation. Until then, the report presents only saved validation results under their true protocols.

References: Research papers / books / websites etc.:-

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